Gavin's Lectures on Physics - Theoretical Mechanics



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Contents

1 The Principle of Least Action (Hamilton's Principle) 2
- 1.1 Action Functional and Variation . . . 2
- 1.2 Relation to Newton's Second Law . . . 2

2 Lagrangian Mechanics and Applications 3
- 2.1 Constraints and Generalized Coordinates . . . 3
  - 2.1.1 Classification by Form of Constraint Equation . . . 3
  - 2.1.2 Classification by Time Dependence . . . 3
  - 2.1.3 Geometric vs. Kinematic Constraints . . . 4
  - 2.1.4 Generalized Coordinates . . . 4
  - 2.1.5 Lagrange Multipliers for Holonomic Constraints . . . 4
- 2.2 Remarks on the Lagrangian Formalism . . . 5
- 2.3 D'Alembert's Principle and Virtual Work . . . 5
- 2.4 Lagrange's Equations from D'Alembert's Principle . . . 5
- 2.5 Small Oscillations (Normal Modes) . . . 6
- 2.6 Rigid Body Rotation . . . 7
- 2.7 Central Force and the Runge–Lenz Vector . . . 8

3 Noether's Theorem and Symmetries 10
- 3.1 Ignorable / Cyclic Coordinate: q_j . . . 10
- 3.2 Noether's Theorem . . . 10
  - 3.2.1 Homogeneity of Space ⇒ Conservation of Momentum . . . 11
  - 3.2.2 Isotropy of Space ⇒ Conservation of Angular Momentum . . . 11
  - 3.2.3 Homogeneity of Time ⇒ Conservation of Energy . . . 12

4 Hamiltonian Mechanics and Applications 13
- 4.1 Hamilton's Canonical Equations . . . 13
- 4.2 Poisson Brackets and Poisson's Theorem . . . 13
- 4.3 Liouville's Theorem . . . 13
- 4.4 Canonical Transformations . . . 14
- 4.5 Hamilton-Jacobi Theory . . . 15
- 4.6 Action-Angle Variables . . . 16
  - 4.6.1 The Simple Harmonic Oscillator . . . 16
  - 4.6.2 Liouville's Theorem on Integrable systems . . . 16
  - 4.6.3 Action-Angle Variables for 1d Systems . . . 16
  - 4.6.4 Action-Angle Variables for the Kepler Problem . . . 17
- 4.7 Adiabatic Invariants . . . 19

1 The Principle of Least Action (Hamilton's Principle)

1.1 Action Functional and Variation

The action for a mechanical system with generalized coordinates $q_\alpha(t)$ is defined as

$$S[q(t)] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, dt,$$

where $L = T - V$ is the Lagrangian. The principle of least action states that the physical trajectory is an extremum of S with respect to variations $\delta q_\alpha(t)$ that vanish at the endpoints:

$$\delta q_\alpha(t_1) = \delta q_\alpha(t_2) = 0.$$

Requiring $\delta S = 0$ yields

$$\delta S = \int_{t_1}^{t_2} \sum_\alpha \left(\frac{\partial L}{\partial q_\alpha} \delta q_\alpha + \frac{\partial L}{\partial \dot{q}_\alpha} \delta \dot{q}_\alpha \right) dt = 0.$$

Integrating the second term by parts and using endpoint conditions, we obtain the Euler–Lagrange equations:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_\alpha} \right) - \frac{\partial L}{\partial q_\alpha} = 0. \tag{1}$$

1.2 Relation to Newton's Second Law

For a single particle in Cartesian coordinates, $L = \frac{1}{2} m \dot{\mathbf{r}}^2 - V(\mathbf{r})$. The Euler–Lagrange equation gives

$$m\ddot{\mathbf{r}} = -\nabla V = \mathbf{F},$$

which is Newton's second law. Thus the variational principle contains Newtonian mechanics as a special case.

2 Lagrangian Mechanics and Applications

2.1 Constraints and Generalized Coordinates

A system of N particles has $3N$ coordinates. In the presence of constraints, the number of independent degrees of freedom is reduced. Constraints can be classified in several ways, and their treatment often involves the introduction of generalized coordinates and, when constraint forces are of interest, the method of Lagrange multipliers.

2.1.1 Classification by Form of Constraint Equation

- **Holonomic constraints**: These are constraints that can be expressed as algebraic equations relating the coordinates and time:

  $$f(\mathbf{r}_1, \ldots, \mathbf{r}_N, t) = 0.$$

  They are integrable and reduce the number of degrees of freedom. Example: a rigid rod connecting two particles imposes $|\mathbf{r}_1 - \mathbf{r}_2| = l$.

- **Non-holonomic constraints**: These cannot be written as algebraic relations among coordinates alone. They often appear as inequalities (e.g., a particle confined to a table, $z \geq 0$) or as non-integrable differential relations. Example: a disk rolling without slipping on a plane involves velocity constraints that cannot be integrated to position constraints.

- **Differential constraints**: A special class of non-holonomic constraints are those given by differential forms:

  $$\sum_{\alpha} a_\alpha(q, t)\, dq_\alpha + a_0(q, t)\, dt = 0.$$

  If this form is integrable (i.e., there exists an integrating factor making it an exact differential), then the constraint is actually holonomic; otherwise, it is non-holonomic.

2.1.2 Classification by Time Dependence

- **Scleronomic (stable) constraints**: Constraints that do not depend explicitly on time:

  $$f(\mathbf{r}_1, \ldots, \mathbf{r}_N) = 0.$$

  The configuration space is fixed. Example: a bead on a rigid circular wire.

- **Rheonomic (unstable) constraints**: Constraints that depend explicitly on time:

  $$f(\mathbf{r}_1, \ldots, \mathbf{r}_N, t) = 0.$$

  The constraints move or change with time. Example: a bead on a circular wire whose radius changes with time.

2.1.3 Geometric vs. Kinematic Constraints

- **Geometric constraints**: Restrict the positions of particles (e.g., a particle on a sphere).
- **Kinematic constraints**: Restrict velocities (e.g., rolling condition $\dot{x} - R\dot{\theta} = 0$). If kinematic constraints are integrable, they reduce to geometric (holonomic) constraints.

2.1.4 Generalized Coordinates

If there are k independent holonomic constraints, the number of independent degrees of freedom is $s = 3N - k$. We introduce generalized coordinates $q_1, \ldots, q_s$ such that

$$\mathbf{r}_i = \mathbf{r}_i(q_1, \ldots, q_s, t).$$

Equivalently, for holonomic constraints $f_\alpha(x_A, t) = 0$ with $\alpha = 1, \ldots, 3N - n$, we can write

$$x_A = x_A\left(q_1, \ldots, q_n\right).$$

Virtual displacements $\delta\mathbf{r}_i$ are defined as infinitesimal changes consistent with the constraints at fixed time:

$$\delta\mathbf{r}_i = \sum_\alpha \frac{\partial \mathbf{r}_i}{\partial q_\alpha} \delta q_\alpha.$$

2.1.5 Lagrange Multipliers for Holonomic Constraints

To find the forces of constraint, we treat holonomic constraints $f_\alpha(x_A, t) = 0$ by introducing multipliers λ_α. The modified Lagrangian is

$$L' = L\left(x^A, \dot{x}^A\right) + \lambda_\alpha f_\alpha\left(x^A, t\right).$$

The constraint equations are obtained by variation with respect to λ_α:

$$\frac{\partial L'}{\partial \lambda_\alpha} = f_\alpha\left(x^A, t\right) = 0,$$

and the equations of motion become

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}^A}\right) - \frac{\partial L}{\partial x^A} = \lambda_\alpha \frac{\partial f_\alpha}{\partial x^A}.$$

In terms of generalized coordinates, we can choose

$$x_A \to \begin{cases} q_i, & (i = 1, \ldots, n), \\ f_\alpha, & (\alpha = 1, \ldots, 3N - n), \end{cases}$$

so that the equations reduce to

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = \lambda_\alpha \frac{\partial f_\alpha}{\partial q_i}. \tag{2}$$

The multipliers λ_α are directly related to the **generalized constraint forces**. Specifically, the generalized force of constraint is given by

$$Q_i^{\text{constraint}} = \sum_\alpha \lambda_\alpha \frac{\partial f_\alpha}{\partial q_i}.$$

For example, in the case of a simple pendulum with constraint $f(r - l) = 0$, this formalism yields the tension in the string.

2.2 Remarks on the Lagrangian Formalism

For a system described by generalized coordinates q_i and velocities $\dot{q}_i$, the Lagrangian is $L = L(q_i, \dot{q}_i, t)$. The Euler–Lagrange equations and the definition of canonical momenta are summarized as

$$\begin{cases} \dfrac{d}{dt}\left(\dfrac{\partial L}{\partial \dot{q}_i}\right) - \dfrac{\partial L}{\partial q_i} = 0, \\ p_i = \dfrac{\partial L}{\partial \dot{q}_i}, \end{cases} \qquad \Rightarrow \quad \dot{p}_i = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = \frac{\partial L}{\partial q_i}.$$

The Lagrangian is not unique: it can be multiplied by a constant (if the action is scaled) or augmented by a total time derivative of any function of coordinates and time. That is, the following transformations leave the equations of motion invariant:

$$L' = \alpha L, \quad (\alpha \in \mathbb{R}), \qquad L' = L + \frac{df}{dt}.$$

2.3 D'Alembert's Principle and Virtual Work

D'Alembert's principle extends the principle of virtual work to dynamics:

$$\sum_i (\mathbf{F}_i - \dot{\mathbf{p}}_i) \cdot \delta \mathbf{r}_i = 0,$$

where $\mathbf{p}_i = m_i \dot{\mathbf{r}}_i$. For ideal constraints (no work done by constraint forces), this reduces to

$$\sum_i \mathbf{F}_i^{\text{applied}} \cdot \delta \mathbf{r}_i = \sum_\alpha Q_\alpha \delta q_\alpha,$$

with generalized forces

$$Q_\alpha = \sum_i \mathbf{F}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_\alpha}.$$

For static equilibrium, $\dot{\mathbf{p}}_i = 0$, so the virtual work vanishes: $\sum_\alpha Q_\alpha \delta q_\alpha = 0$, hence $Q_\alpha = 0$ for all α.

2.4 Lagrange's Equations from D'Alembert's Principle

D'Alembert's principle for a system of N particles with ideal constraints is

$$\sum_i (\mathbf{F}_i - m_i \ddot{\mathbf{r}}_i) \cdot \delta \mathbf{r}_i = 0,$$

where $\delta \mathbf{r}_i = \sum_\alpha \frac{\partial \mathbf{r}_i}{\partial q_\alpha} \delta q_\alpha$ are virtual displacements. Define generalized forces

$$Q_\alpha = \sum_i \mathbf{F}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_\alpha}.$$

Using the kinetic energy $T = \frac{1}{2} \sum_i m_i \dot{\mathbf{r}}_i^2$, one can show (by expressing $\ddot{\mathbf{r}}_i \cdot \delta \mathbf{r}_i$ in terms of T and its derivatives) that

$$\sum_i m_i \ddot{\mathbf{r}}_i \cdot \delta \mathbf{r}_i = \sum_\alpha \left[\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_\alpha}\right) - \frac{\partial T}{\partial q_\alpha} \right] \delta q_\alpha.$$

Substituting into D'Alembert's principle yields

$$\sum_{\alpha} \left[Q_\alpha - \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_\alpha}\right) + \frac{\partial T}{\partial q_\alpha} \right] \delta q_\alpha = 0.$$

Since the δq_α are independent, we obtain Lagrange's equations of the first kind:

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_\alpha}\right) - \frac{\partial T}{\partial q_\alpha} = Q_\alpha.$$

For Conservative Forces $Q_\alpha = -\partial V/\partial q_\alpha$, and with $L = T - V$, this reduces to the standard Euler–Lagrange equations (1).

For Non-Conservative Forces When non-conservative forces (e.g., friction) are present, we write

$$Q_\alpha = -\frac{\partial V}{\partial q_\alpha} + Q_\alpha^{\text{nc}}.$$

If the dissipative forces are linear in velocities, we introduce the Rayleigh dissipation function

$$\mathcal{F} = \frac{1}{2}\sum_{i,j} c_{ij}\dot{q}_i\dot{q}_j,$$

where the coefficient matrix c_{ij} is taken to be symmetric. Then the generalized non-conservative forces are

$$Q_\alpha^{\text{nc}} = -\frac{\partial \mathcal{F}}{\partial \dot{q}_\alpha} = \sum_i c_{\alpha i}\dot{q}_i.$$

The modified Lagrange equations become

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_\alpha}\right) - \frac{\partial L}{\partial q_\alpha} = Q_\alpha^{\text{nc}} = -\frac{\partial \mathcal{F}}{\partial \dot{q}_\alpha}. \tag{3}$$

2.5 Small Oscillations (Normal Modes)

Near stable equilibrium q_α^0, let $\eta_\alpha = q_\alpha - q_\alpha^0$. Expand the potential and kinetic energy to quadratic order:

$$V = V_0 + \frac{1}{2}\sum_{\alpha,\beta} K_{\alpha\beta}\eta_\alpha\eta_\beta, \qquad T = \frac{1}{2}\sum_{\alpha,\beta} M_{\alpha\beta}\dot{\eta}_\alpha\dot{\eta}_\beta,$$

where

$$K_{\alpha\beta} = \left.\frac{\partial^2 V}{\partial q_\alpha \partial q_\beta}\right|_0$$

and $M_{\alpha\beta}$ is the mass matrix. The Lagrangian is

$$L = \frac{1}{2}\sum_{\alpha,\beta}\left(M_{\alpha\beta}\dot{\eta}_\alpha\dot{\eta}_\beta - K_{\alpha\beta}\eta_\alpha\eta_\beta\right).$$

The equations of motion yield the generalized eigenvalue problem

$$\sum_\beta \left(K_{\alpha\beta} - \omega^2 M_{\alpha\beta}\right) A_\beta = 0. \tag{4}$$

The s eigenfrequencies ω_k determine the normal modes. In terms of normal coordinates ξ_k, the Lagrangian decouples:

$$L = \frac{1}{2}\sum_k \left(\dot{\xi}_k^2 - \omega_k^2 \xi_k^2\right).$$

2.6 Rigid Body Rotation

For a rigid body, the kinetic energy separates into translational and rotational parts via König's theorem:

$$T = \frac{1}{2}MV_{\rm cm}^2 + \frac{1}{2}\boldsymbol{\omega}\cdot\boldsymbol{I}\cdot\boldsymbol{\omega}. \tag{5}$$

Inertia tensor and angular momentum For rigid rotation, $\dot{\boldsymbol{r}}_i' = \boldsymbol{\omega}\times\boldsymbol{r}_i'$ The angular momentum is

$$\begin{aligned}\boldsymbol{L} &= \sum_i \boldsymbol{r}_i' \times m_i\dot{\boldsymbol{r}}_i' \\ &= \sum_i m_i \boldsymbol{r}_i' \times (\boldsymbol{\omega}\times\boldsymbol{r}_i') \\ &= \sum_i m_i \left[\boldsymbol{\omega}\, r_i'^2 - \boldsymbol{r}_i'(\boldsymbol{r}_i'\cdot\boldsymbol{\omega})\right] \\ &= \boldsymbol{I}\cdot\boldsymbol{\omega}.\end{aligned}$$

In component notation:

$$L_\alpha = \sum_\beta I_{\alpha\beta}\omega_\beta.$$

The inertia tensor components are explicitly:

$$I_{xx} = \int (y^2+z^2)\,dm, \qquad I_{yy} = \int (z^2+x^2)\,dm, \qquad I_{zz} = \int (x^2+y^2)\,dm,$$

$$I_{xy} = I_{yx} = -\int xy\,dm, \qquad I_{yz} = I_{zy} = -\int yz\,dm, \qquad I_{zx} = I_{xz} = -\int xz\,dm.$$

In the principal axis frame, the tensor is diagonal with moments I_1, I_2, I_3, and

$$T_{\rm rot} = \frac{1}{2}\left(I_1\omega_1^2 + I_2\omega_2^2 + I_3\omega_3^2\right).$$

Euler's equations for torque-free motion In the inertial frame, $d\boldsymbol{L}/dt\big|_{\rm inertial} = \boldsymbol{M}_{\rm ext} = 0$. The time derivative rule for a rotating frame gives

$$\left.\frac{d\boldsymbol{L}}{dt}\right|_{\rm inertial} = \left.\frac{d\boldsymbol{L}}{dt}\right|_{\rm rot} + \boldsymbol{\omega}\times\boldsymbol{L} = 0.$$

In principal axes, $\boldsymbol{L} = (I_1\omega_1, I_2\omega_2, I_3\omega_3)$. The cross product is

$$\boldsymbol{\omega}\times\boldsymbol{L} = \big((I_3-I_2)\omega_2\omega_3,\ (I_1-I_3)\omega_3\omega_1,\ (I_2-I_1)\omega_1\omega_2\big).$$

Separating components yields Euler's equations:

$$\begin{aligned} I_1\dot{\omega}_1 + (I_3 - I_2)\omega_2\omega_3 &= 0, \\ I_2\dot{\omega}_2 + (I_1 - I_3)\omega_3\omega_1 &= 0, \\ I_3\dot{\omega}_3 + (I_2 - I_1)\omega_1\omega_2 &= 0. \end{aligned} \tag{6}$$

The Euler angles (ϕ, θ, ψ) describe the orientation; their time derivatives relate to $\boldsymbol{\omega}$ via standard formulas.

2.7 Central Force and the Runge–Lenz Vector

Effective radial potential For a particle in a central potential $V(r)$, the angular momentum is conserved. In polar coordinates,

$$T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2).$$

Since θ is cyclic, $p_\theta = mr^2\dot{\theta} = L$ is conserved, so $\dot{\theta} = L/(mr^2)$. The total energy is

$$E = \frac{1}{2}m\dot{r}^2 + \frac{L^2}{2mr^2} + V(r).$$

Thus the effective radial motion is governed by the effective potential:

$$V_{\text{eff}}(r) = V(r) + \frac{L^2}{2mr^2}. \tag{7}$$

Binet's orbit equation To eliminate time and directly solve the orbital shape $r(\theta)$, we use the radial force equation:

$$m(\ddot{r} - r\dot{\theta}^2) = F(r).$$

Change variable $u = 1/r$. We have

$$\dot{r} = \frac{dr}{d\theta}\dot{\theta} = -\frac{1}{u^2}\frac{du}{d\theta}\cdot\frac{Lu^2}{m} = -\frac{L}{m}\frac{du}{d\theta},$$

$$\ddot{r} = \frac{d\dot{r}}{d\theta}\dot{\theta} = -\frac{L^2u^2}{m^2}\frac{d^2u}{d\theta^2}, \qquad r\dot{\theta}^2 = \frac{L^2u^3}{m^2}.$$

Substituting into the radial equation gives Binet's equation:

$$\frac{d^2u}{d\theta^2} + u = -\frac{m}{L^2u^2}F\left(\frac{1}{u}\right). \tag{8}$$

Runge–Lenz vector for the Kepler problem For the Kepler potential $V(r) = -\alpha/r$, the Runge–Lenz vector is defined as

$$\boldsymbol{A} = \boldsymbol{p} \times \boldsymbol{L} - m\alpha\frac{\boldsymbol{r}}{r}. \tag{9}$$

To prove its conservation, differentiate with respect to time:

$$\frac{d\boldsymbol{A}}{dt} = \dot{\boldsymbol{p}} \times \boldsymbol{L} + \boldsymbol{p} \times \dot{\boldsymbol{L}} - m\alpha\frac{d}{dt}\left(\frac{\boldsymbol{r}}{r}\right).$$

For a central force, $\dot{\boldsymbol{L}} = 0$, so the second term vanishes. Newton's law gives

$$\dot{\boldsymbol{p}} = \boldsymbol{F} = -\frac{\alpha}{r^3}\boldsymbol{r}.$$

Thus

$$\dot{\boldsymbol{p}} \times \boldsymbol{L} = -\frac{\alpha}{r^3}\boldsymbol{r} \times (\boldsymbol{r} \times \boldsymbol{p}) = \alpha\left(\frac{\boldsymbol{p}}{r} - \frac{\boldsymbol{r}(\boldsymbol{r}\cdot\boldsymbol{p})}{r^3}\right).$$

The derivative of the unit vector is

$$\frac{d}{dt}\left(\frac{\boldsymbol{r}}{r}\right) = \frac{\boldsymbol{p}}{mr} - \frac{\boldsymbol{r}(\boldsymbol{r}\cdot\boldsymbol{p})}{mr^3}.$$

Therefore

$$-m\alpha\frac{d}{dt}\left(\frac{\boldsymbol{r}}{r}\right) = \alpha\left(-\frac{\boldsymbol{p}}{r} + \frac{\boldsymbol{r}(\boldsymbol{r}\cdot\boldsymbol{p})}{r^3}\right).$$

Summing all terms yields complete cancellation, hence

$$\frac{d\boldsymbol{A}}{dt} = 0.$$

The vector $\boldsymbol{A}$ points from the focus to the perihelion, and its magnitude is proportional to the orbital eccentricity.

3 Noether's Theorem and Symmetries

Noether's theorem establishes a profound connection between symmetries of a physical system and its conserved quantities. For a Lagrangian system, any continuous symmetry of the action (or of the Lagrangian) implies the existence of a corresponding constant of motion. Below we formalize this idea and illustrate its most important applications.

A function $F(q_i, \dot{q}_i, t)$ is a constant of motion (conserved quantity) if

$$0 = \frac{dF}{dt} = \sum_{j=1}^{n} \left(\frac{\partial F}{\partial q_j} \dot{q}_j + \frac{\partial F}{\partial \dot{q}_j} \ddot{q}_j \right) + \frac{\partial F}{\partial t}.$$

The Hamiltonian is defined as

$$H = \sum_j \dot{q}_j \frac{\partial L}{\partial \dot{q}_j} - L.$$

If $\partial L/\partial t = 0$, then

$$\frac{dH}{dt} = \sum_j \left[\ddot{q}_j \frac{\partial L}{\partial \dot{q}_j} + \dot{q}_j \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} \dot{q}_j - \frac{\partial L}{\partial \dot{q}_j} \ddot{q}_j \right] = 0.$$

3.1 Ignorable / Cyclic Coordinate: q_j

If a coordinate q_j does not appear explicitly in the Lagrangian,

$$\begin{cases} \dfrac{\partial L}{\partial q_j} = 0, \\ p_j = \dfrac{\partial L}{\partial \dot{q}_j}, \end{cases} \qquad \Rightarrow \quad \frac{d}{dt} p_j = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) = \frac{\partial L}{\partial q_j} = 0.$$

Thus, a coordinate that does not appear explicitly in the Lagrangian (a cyclic coordinate) gives rise to a conserved canonical momentum.

3.2 Noether's Theorem

Consider a one-parameter family of transformations:

$$\begin{cases} q_i(t) \to Q_i(s, t), \quad s \in \mathbb{R}, \\ Q_i(s = 0, t) = q_i(t). \end{cases}$$

If the transformation leaves the Lagrangian invariant (up to a total time derivative), i.e.,

$$\frac{\partial}{\partial s} L\Big(Q_i(s, t), \dot{Q}_i(s, t), t \Big) = 0,$$

then this transformation is a continuous symmetry of the Lagrangian L, and there exists a conserved quantity F with $dF/dt = 0$.

Proof: Evaluating the derivative at $s = 0$:

$$\begin{aligned}
0 &= \left.\frac{\partial L}{\partial s}\right|_{s=0} \\
&= \left[\frac{\partial L}{\partial Q_i}\frac{\partial Q_i}{\partial s} + \frac{\partial L}{\partial \dot{Q}_i}\frac{\partial \dot{Q}_i}{\partial s}\right]_{s=0} \\
&= \left.\frac{\partial L}{\partial q_i}\frac{\partial Q_i}{\partial s}\right|_{s=0} + \left.\frac{\partial L}{\partial \dot{q}_i}\frac{\partial \dot{Q}_i}{\partial s}\right|_{s=0} \\
&= \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right)\left.\frac{\partial Q_i}{\partial s}\right|_{s=0} + \frac{\partial L}{\partial \dot{q}_i}\frac{d}{dt}\left(\left.\frac{\partial Q_i}{\partial s}\right|_{s=0}\right) \\
&= \frac{d}{dt}\left(\left.\frac{\partial L}{\partial \dot{q}_i}\frac{\partial Q_i}{\partial s}\right|_{s=0}\right).
\end{aligned}$$

Hence the conserved quantity is

$$F = \left.\frac{\partial L}{\partial \dot{q}_i}\frac{\partial Q_i}{\partial s}\right|_{s=0}, \qquad \frac{dF}{dt} = 0. \tag{10}$$

3.2.1 Homogeneity of Space ⇒ Conservation of Momentum

For a free particle,

$$L = \frac{1}{2}m\dot{\mathbf{r}}^2 - V(\mathbf{r}).$$

Under a spatial translation $\mathbf{r} \to \mathbf{r}' = \mathbf{r} + s\hat{n}$, the Lagrangian is invariant. Then

$$0 = \frac{\partial L}{\partial s} = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\mathbf{r}}}\cdot\hat{n}\right) = \frac{d}{dt}\left(m\dot{\mathbf{r}}\cdot\hat{n}\right) = \frac{d}{dt}\left(\mathbf{p}\cdot\hat{n}\right).$$

Since $\hat{n}$ is arbitrary,

$$\Rightarrow \frac{d\mathbf{p}}{dt} = 0.$$

3.2.2 Isotropy of Space ⇒ Conservation of Angular Momentum

Under an infinitesimal rotation $\mathbf{r} \to \mathbf{r}' = \mathbf{r} + \alpha\left(\hat{n}\times\mathbf{r}\right)$ (with α the parameter), invariance of the Lagrangian gives

$$\begin{aligned}
0 &= \frac{\partial L}{\partial \alpha} \\
&= \frac{d}{dt}\left[\frac{\partial L}{\partial \dot{\mathbf{r}}}\cdot\left(\hat{n}\times\mathbf{r}\right)\right] \\
&= \frac{d}{dt}\left[\mathbf{p}\cdot\left(\hat{n}\times\mathbf{r}\right)\right] \\
&= \frac{d}{dt}\left[\hat{n}\cdot\left(\mathbf{r}\times\mathbf{p}\right)\right] \\
&= \frac{d}{dt}\left(\hat{n}\cdot\mathbf{L}\right).
\end{aligned}$$

Since $\hat{n}$ is arbitrary,

$$\Rightarrow \frac{d\mathbf{L}}{dt} = 0.$$

3.2.3 Homogeneity of Time ⇒ Conservation of Energy

If the Lagrangian does not depend explicitly on time, i.e., $\partial L/\partial t = 0$, then time translation $t \to t' = t + s$ is a symmetry. This implies

$$\frac{\partial L}{\partial t} = 0 \quad \Rightarrow \quad \frac{dH}{dt} = \frac{d}{dt}\left[\sum_i \dot{q}_i \left(\frac{\partial L}{\partial \dot{q}_i}\right) - L\right] = 0,$$

so the Hamiltonian (energy) is conserved.

4 Hamiltonian Mechanics and Applications

4.1 Hamilton's Canonical Equations

Given the Lagrangian $L(q, \dot{q}, t)$, define canonical momenta

$$p_\alpha = \frac{\partial L}{\partial \dot{q}_\alpha}.$$

The Hamiltonian is obtained by Legendre transformation:

$$H(q, p, t) = \sum_\alpha p_\alpha \dot{q}_\alpha - L.$$

Then the equations of motion take the symmetric form

$$\dot{q}_\alpha = \frac{\partial H}{\partial p_\alpha}, \qquad \dot{p}_\alpha = -\frac{\partial H}{\partial q_\alpha}. \tag{11}$$

A coordinate q_α that does not appear explicitly in H is cyclic; its conjugate momentum p_α is conserved.

4.2 Poisson Brackets and Poisson's Theorem

For any two functions $f(q, p, t)$ and $g(q, p, t)$, their Poisson bracket is defined as

$$\{f, g\} = \sum_{\alpha=1}^{s} \left(\frac{\partial f}{\partial q_\alpha}\frac{\partial g}{\partial p_\alpha} - \frac{\partial f}{\partial p_\alpha}\frac{\partial g}{\partial q_\alpha} \right).$$

The time evolution of any function f is given by

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \{f, H\}. \tag{12}$$

Poisson's theorem states that if f and g are constants of motion (i.e., their time derivatives vanish) **and** they do not depend explicitly on time ($\partial f/\partial t = \partial g/\partial t = 0$), then their Poisson bracket $\{f, g\}$ is also a constant of motion. This property is essential for integrability.

4.3 Liouville's Theorem

The volume of a region in phase space is invariant under Hamiltonian flow:

$$V(R) = \int_R dq_1 \cdots dq_s \, dp_1 \cdots dp_s.$$

Proof: The phase-space flow velocity is

$$\boldsymbol{v} = (\dot{q}_1, \ldots, \dot{q}_s, \dot{p}_1, \ldots, \dot{p}_s).$$

Its divergence is

$$\nabla \cdot \boldsymbol{v} = \sum_{\alpha=1}^{s} \left(\frac{\partial \dot{q}_\alpha}{\partial q_\alpha} + \frac{\partial \dot{p}_\alpha}{\partial p_\alpha} \right).$$

Using Hamilton's canonical equations,

$$\dot{q}_\alpha = \frac{\partial H}{\partial p_\alpha}, \qquad \dot{p}_\alpha = -\frac{\partial H}{\partial q_\alpha},$$

we have

$$\frac{\partial \dot{q}_\alpha}{\partial q_\alpha} = \frac{\partial^2 H}{\partial q_\alpha \partial p_\alpha}, \qquad \frac{\partial \dot{p}_\alpha}{\partial p_\alpha} = -\frac{\partial^2 H}{\partial p_\alpha \partial q_\alpha}.$$

Since mixed partial derivatives commute, each pair cancels:

$$\frac{\partial \dot{q}_\alpha}{\partial q_\alpha} + \frac{\partial \dot{p}_\alpha}{\partial p_\alpha} = 0 \quad \Rightarrow \quad \nabla \cdot \boldsymbol{v} = 0.$$

By Reynolds transport theorem,

$$\frac{dV}{dt} = \int_R \nabla \cdot \boldsymbol{v} \, d\Gamma = 0.$$

Thus the Hamiltonian flow is incompressible in phase space, and the density of representative points is constant along trajectories.

4.4 Canonical Transformations

A transformation $(q, p) \to (Q, P)$ is canonical if it preserves the form of Hamilton's equations. The fundamental condition for a canonical transformation is that there exists a generating function F such that

$$\sum_\alpha p_\alpha dq_\alpha - H dt = \sum_\alpha P_\alpha dQ_\alpha - K dt + dF. \tag{13}$$

Type-1 generating function $F_1(q, Q, t)$ For a generating function depending on the old coordinates q and new coordinates Q, the total differential is

$$dF_1 = \sum_\alpha \frac{\partial F_1}{\partial q_\alpha} dq_\alpha + \sum_\alpha \frac{\partial F_1}{\partial Q_\alpha} dQ_\alpha + \frac{\partial F_1}{\partial t} dt.$$

Substituting into the fundamental condition and grouping independent differentials dq_α, dQ_α, and dt, their coefficients must vanish individually. This yields the transformation rules:

$$p_\alpha = \frac{\partial F_1}{\partial q_\alpha}, \qquad P_\alpha = -\frac{\partial F_1}{\partial Q_\alpha}, \qquad K = H + \frac{\partial F_1}{\partial t}.$$

Type-2 generating function $F_2(q, P, t)$ A Legendre transform from F_1 gives $F_2 = F_1 + \sum P_\alpha Q_\alpha$. The transformation rules become

$$p_\alpha = \frac{\partial F_2}{\partial q_\alpha}, \qquad Q_\alpha = \frac{\partial F_2}{\partial P_\alpha}, \qquad K = H + \frac{\partial F_2}{\partial t}.$$

This form is particularly useful for Hamilton-Jacobi theory.

Type-3 generating function $F_3(p, Q, t)$ A Legendre transform from F_1 gives $F_3 = F_1 - \sum p_\alpha q_\alpha$. The total differential is

$$dF_3 = -\sum_\alpha q_\alpha dp_\alpha - \sum_\alpha P_\alpha dQ_\alpha + (K - H)dt = \sum_\alpha \frac{\partial F_3}{\partial p_\alpha} dp_\alpha + \sum_\alpha \frac{\partial F_3}{\partial Q_\alpha} dQ_\alpha + \frac{\partial F_3}{\partial t} dt.$$

Thus the transformation rules are

$$q_\alpha = -\frac{\partial F_3}{\partial p_\alpha}, \qquad P_\alpha = -\frac{\partial F_3}{\partial Q_\alpha}, \qquad K = H + \frac{\partial F_3}{\partial t}.$$

Type-4 generating function $F_4(p, P, t)$ Similarly, $F_4 = F_1 - \sum p_\alpha q_\alpha + \sum P_\alpha Q_\alpha$ gives

$$dF_4 = -\sum_\alpha q_\alpha dp_\alpha + \sum_\alpha Q_\alpha dP_\alpha + (K - H)dt = \sum_\alpha \frac{\partial F_4}{\partial p_\alpha} dp_\alpha + \sum_\alpha \frac{\partial F_4}{\partial P_\alpha} dP_\alpha + \frac{\partial F_4}{\partial t} dt,$$

yielding

$$q_\alpha = -\frac{\partial F_4}{\partial p_\alpha}, \qquad Q_\alpha = \frac{\partial F_4}{\partial P_\alpha}, \qquad K = H + \frac{\partial F_4}{\partial t}.$$

4.5 Hamilton-Jacobi Theory

If we find a canonical transformation such that the new Hamiltonian $K \equiv 0$, then the new coordinates and momenta are constants of motion. The type-2 generating function $S(q, P, t)$ satisfies the Hamilton-Jacobi equation:

$$\frac{\partial S}{\partial t} + H\left(q, \frac{\partial S}{\partial q}, t\right) = 0. \tag{14}$$

Derivation and physical interpretation We seek a canonical transformation with new Hamiltonian $K = 0$. Using a type-2 generating function $S(q, \alpha, t)$, where $P_\alpha = \alpha_\alpha$ are constants, the canonical transformation rule gives $K = H + \partial S/\partial t$. Setting $K = 0$ and substituting $p_\alpha = \partial S/\partial q_\alpha$ into H directly produces the Hamilton-Jacobi equation.

To verify that S is the classical action, compute its total time derivative:

$$\frac{dS}{dt} = \sum \frac{\partial S}{\partial q_\alpha} \dot{q}_\alpha + \frac{\partial S}{\partial t} = \sum p_\alpha \dot{q}_\alpha - H = L.$$

Integrating gives $S = \int L\, dt$, confirming that S is indeed the action functional.

A complete solution $S(q, \alpha, t)$ with s constants α_α (where $\alpha_\alpha = P_\alpha$) gives the trajectories through

$$p_\alpha = \frac{\partial S}{\partial q_\alpha}, \qquad \beta_\alpha = \frac{\partial S}{\partial \alpha_\alpha},$$

where β_α are new constants (the new coordinates Q_α).

Time-independent Hamilton-Jacobi equation For time-independent Hamiltonians ($\partial H/\partial t = 0$), we separate variables:

$$S = W(q, \alpha) - Et.$$

Substituting into the Hamilton-Jacobi equation yields the time-independent equation:

$$H\left(q, \frac{\partial W}{\partial q}\right) = E,$$

where W is Hamilton's characteristic function.

4.6 Action-Angle Variables

4.6.1 The Simple Harmonic Oscillator

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2 \tag{1}$$

$$\Rightarrow \begin{cases} \dot{q} = \dfrac{\partial H}{\partial p} = \dfrac{p}{m}, \\ \dot{p} = -\dfrac{\partial H}{\partial q} = -m\omega^2 q \end{cases} \tag{2}$$

Change variable $(q, p) \to (\theta, I)$:

$$\begin{cases} q = \sqrt{\dfrac{2I}{m\omega}}\ \sin\theta, \\ p = \sqrt{2Im\omega}\ \cos\theta \end{cases} \tag{3}$$

$$H = \omega I \tag{4}$$

$$\begin{cases} \dot{\theta} = \dfrac{\partial H}{\partial I} = \omega, \\ \dot{I} = -\dfrac{\partial H}{\partial \theta} = 0 \end{cases} \tag{5}$$

4.6.2 Liouville's Theorem on Integrable systems

If we can find n mutually Poisson commuting constants of motion $I_1, I_2, \cdots, I_n$, then this implies the existence of angle-action variables (θ_i, I_i), and the system is integrable.

$$\{I_i, I_j\} = 0 \tag{6}$$

4.6.3 Action-Angle Variables for 1d Systems

$$H = \frac{p^2}{2m} + V(q), \qquad H = H(I) = E \tag{7}$$

Choose I as the phase integral:

$$I = \frac{1}{2\pi}\oint p\, dq \tag{8}$$

$$\begin{cases} \dfrac{dI}{dE} = \dfrac{1}{2\pi}\dfrac{d}{dE}\oint p\, dq, \\ E = \dfrac{p^2}{2m} + V(q), \\ p = m\dfrac{dq}{dt} \end{cases} \tag{9}$$

$$\Rightarrow \ \frac{dI}{dE} = \frac{1}{2\pi}\oint dt = \frac{1}{2\pi}\cdot T = \frac{1}{\omega} \tag{10}$$

$$\Rightarrow \begin{cases} \dot{\theta} = \dfrac{\partial H}{\partial I} = \left(\dfrac{dI}{dE}\right)^{-1} = \omega, \\ \dot{I} = -\dfrac{\partial H}{\partial \theta} = 0 \end{cases} \tag{11}$$

$$t = \frac{d}{dE}\int p\,dq \tag{12}$$

$$\theta = \omega t = \frac{dE}{dI}\cdot\frac{d}{dE}\int p\,dq = \frac{d}{dI}\int p\,dq \tag{13}$$

$$\Rightarrow \begin{cases} \theta = \dfrac{d}{dI}\int p\,dq, \\ I = \dfrac{1}{2\pi}\oint p\,dq \end{cases} \tag{14}$$

For higher dimensions, assuming separability, one usually defines each action as the line integral over the corresponding closed orbit:

$$I_i = \frac{1}{2\pi}\oint_{\Gamma_i} p_i\,dq_i \tag{15}$$

where Γ_i are the periods of the invariant tori. In more invariant form:

$$\begin{cases} \theta_i = \dfrac{\partial}{\partial I_i}\int_{\Gamma_i}\sum_j p_j\,dq_j, \\ I_i = \dfrac{1}{2\pi}\oint_{\Gamma_i}\sum_j p_j\,dq_j \end{cases} \tag{16}$$

with Γ_i being the periods of the invariant tori.

4.6.4 Action-Angle Variables for the Kepler Problem

$$\begin{cases} H = E = \dfrac{1}{2}m\dot{r}^2 + \dfrac{1}{2}mr^2\dot{\phi}^2 - \dfrac{G\,Mm}{r}, \\ k = G\,Mm, \\ p_r = m\dot{r}, \qquad p_\phi = L_\phi = mr^2\dot{\phi} \end{cases} \tag{17}$$

$$\Rightarrow\ H = E = \frac{p_r^2}{2m} + \frac{p_\phi^2}{2mr^2} - \frac{k}{r} \tag{18}$$

For ϕ

$$\begin{cases} \dot{\phi} = \dfrac{\partial H}{\partial p_\phi} = \dfrac{p_\phi}{mr^2}, \\ \dot{p}_\phi = -\dfrac{\partial H}{\partial \phi} = 0 \end{cases} \tag{19}$$

$$I_\phi = \frac{1}{2\pi}\oint p_\phi\,d\phi = p_\phi\cdot\frac{1}{2\pi}\oint d\phi = p_\phi \tag{20}$$

$$\begin{cases} \dot{\phi} = \dfrac{I_\phi}{mr^2}, \\ \dot{I}_\phi = 0 \end{cases} \tag{21}$$

For r

$$\begin{cases} \dot{r} = \dfrac{\partial H}{\partial p_r} = \dfrac{p_r}{m}, \\ \dot{p}_r = -\dfrac{\partial H}{\partial r} = -\dfrac{p_\phi^2}{mr^3} + \dfrac{k}{r^2} \end{cases} \tag{22}$$

The action for the radial motion is

$$I_r = \frac{1}{2\pi} \oint p_r \, dr = \frac{1}{\pi} \int_{r_{min}}^{r_{max}} p_r \, dr \tag{23}$$

with

$$E = \frac{p_r^2}{2m} + \frac{I_\phi^2}{2mr^2} - \frac{k}{r} = -|E| \tag{24}$$

Thus

$$p_r = \frac{\sqrt{2m|E|}}{r} \sqrt{-r^2 + \frac{k}{|E|} r - \frac{I_\phi^2}{2m|E|}} \tag{25}$$

and consequently

$$I_r = \frac{1}{\pi} \int_{r_{min}}^{r_{max}} \frac{\sqrt{2m|E|}}{r} \cdot \sqrt{-r^2 + \frac{k}{|E|} r - \frac{I_\phi^2}{2m|E|}} \, dr \tag{26}$$

The turning points satisfy

$$\begin{cases} r_{max} \cdot r_{min} = \dfrac{I_\phi^2}{2m|E|}, \\ r_{min} + r_{max} = \dfrac{k}{|E|} \end{cases} \tag{27}$$

The integral evaluates to

$$I_r = \frac{\sqrt{2m|E|}}{\pi} \int_{r_{min}}^{r_{max}} \frac{1}{r} \sqrt{(r_{max} - r)(r - r_{min})} \, dr \tag{28}$$

$$= \frac{\sqrt{2m|E|}}{\pi} \left[\frac{\pi}{2} \big(r_{min} + r_{max} \big) - \pi \sqrt{r_{min} r_{max}} \right] \tag{29}$$

$$= \frac{k}{2} \sqrt{\frac{2m}{|E|}} - I_\phi \tag{30}$$

Solving for the energy:

$$H = E = -|E| = -\frac{mk^2}{2(I_r + I_\phi)^2} \tag{31}$$

The frequencies are given by

$$\begin{cases} \dot{\theta}_r = \dfrac{\partial H}{\partial I_r}, \\ \dot{I}_r = -\dfrac{\partial H}{\partial \theta_r} = 0 \end{cases} \tag{32}$$

and since

$$H(I_r, I_\phi) = -\frac{mk^2}{2(I_r + I_\phi)^2} \tag{33}$$

is symmetric in I_r and I_ϕ, the motion is degenerate.

4.7 Adiabatic Invariants

Consider a system whose Hamiltonian depends on a slowly varying parameter $\lambda(t)$:

$$H = H(q, p, \lambda(t)).$$

Let τ be the timescale of variation of λ and T the period of the system. The adiabatic condition is $\tau \gg T$. Under this condition, the action variable

$$I = \frac{1}{2\pi} \oint p\, dq$$

is approximately conserved, i.e., it is an adiabatic invariant.

To show this, define

$$I = I(E, \lambda) = \frac{1}{2\pi} \oint p\, dq.$$

Then

$$\dot{I} = \left.\frac{\partial I}{\partial E}\right|_{\lambda} \dot{E} + \left.\frac{\partial I}{\partial \lambda}\right|_{E} \dot{\lambda}.$$

Using

$$\left.\frac{\partial I}{\partial E}\right|_{\lambda} = \frac{T}{2\pi}, \qquad \left.\frac{\partial I}{\partial \lambda}\right|_{E} = -\frac{1}{2\pi} \int_0^T \left.\frac{\partial H}{\partial \lambda}\right|_{E} dt',$$

and the equations of motion, one obtains

$$\dot{I} = \frac{\dot{\lambda}}{2\pi} \left[T \left.\frac{\partial H}{\partial \lambda}\right|_{E} - \int_0^T \left.\frac{\partial H}{\partial \lambda}\right|_{E} dt' \right].$$

Averaging over one fast period gives

$$\langle \dot{I} \rangle = 0,$$

so the action is conserved on average. Thus the action variable is an adiabatic invariant.

The Bohr-Sommerfeld quantization condition states that for integrable systems,

$$I = \frac{1}{2\pi} \oint p\, dq = n\hbar, \qquad n \in \mathbb{Z}.$$

This quantization rule played a crucial role in the old quantum theory and remains a cornerstone of semiclassical physics.